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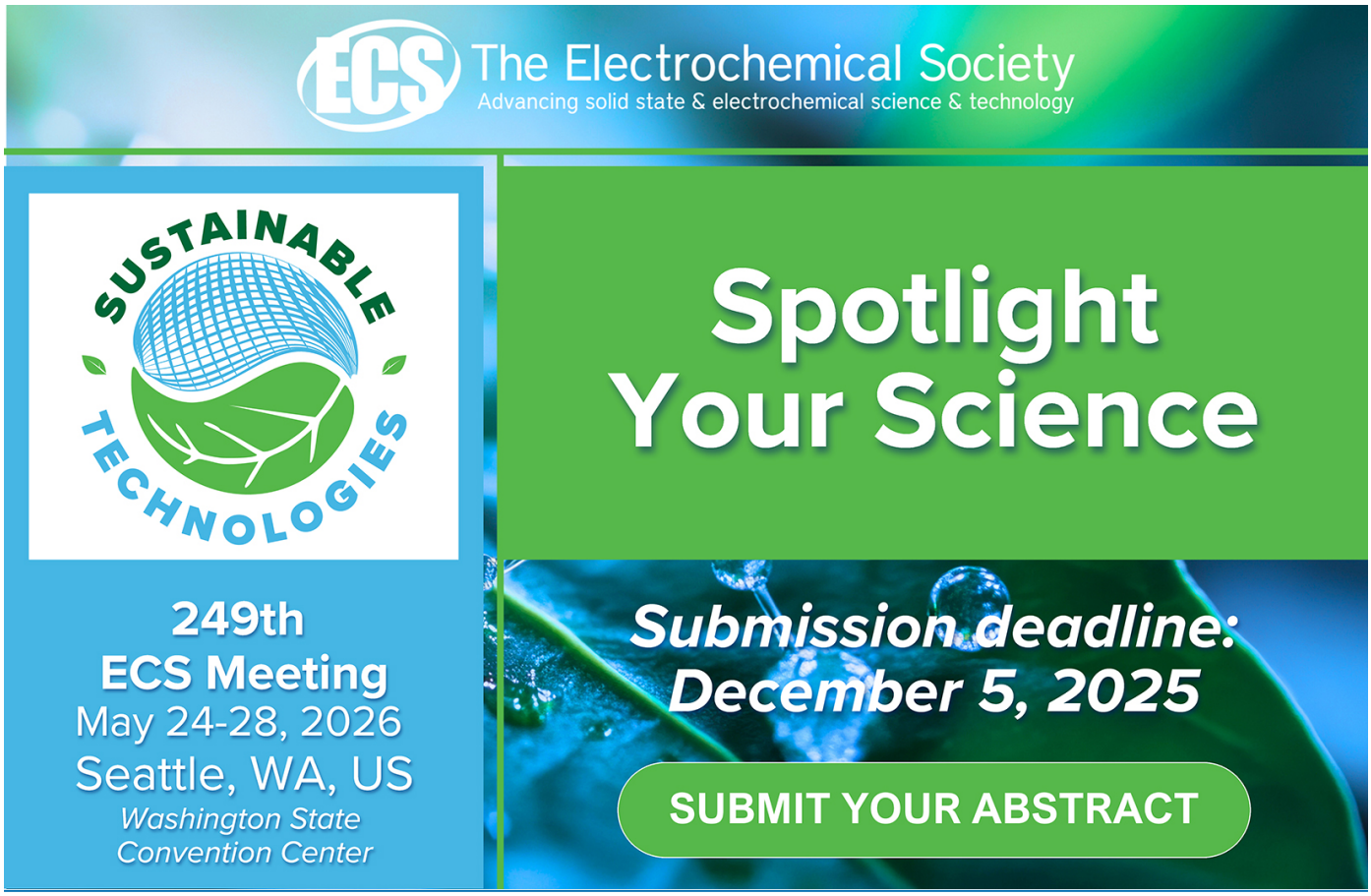

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Robotic Arm Obstacle Avoidance Path Planning Based on Improved RRT algorithm

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Abstract. In order to solve the problems of low sampling efficiency, poor adaptability to fixed-step-size environments, and poor path quality of the traditional Rapidly-Exploring Random Tree (RRT) algorithm in three-dimensional environments, an enhanced RRT algorithm is put forth. Firstly, a differential sampling mechanism is suggested as a solution to the RRT algorithm's poor sampling efficiency issue by minimizing the number of times spatial regions are sampled; secondly, an adaptive target bias and adaptive step size mechanism is proposed to adaptively adjust the target bias probability and sampling step size in combination with the spatial occupancy of the obstacles and the progress of the path planning, so as to solve the problems of poor adaptability of the fixed target bias probability and the fixed-step-size environment, to Further improve the path planning efficiency; finally, using greedy pruning strategy and B spline curve optimization strategy, the initial path is pruned and optimized, which removes the redundant nodes, smoothes the path, and improves the path quality; simulation experiments are analyzed in three-dimensional obstacle environments, and the outcomes demonstrate that the improved RRT algorithm in this work can successfully raise the quality of the path and search efficiency. The improved algorithm is applied to the robotic arm physical platform. The robotic arm can move smoothly, accurately, and quickly to the desired location., which further demonstrates the algorithm's viability and efficacy.

Keywords: Robotic Arm, Path Planning, Differential Sampling, Adaptive Target Bias, Adaptive Step Size, Greedy Pruning

1. Introduction

Given the quick development of industrial automation and intelligent manufacturing, robotic arms are increasingly used as core execution equipment in various fields [1]. However, the efficient and precise operation of a robotic arm is highly dependent on its path planning capability [2]. Finding a collision-free motion trajectory for a robotic arm from its starting point to its destination in the workplace is the goal of path planning [3-4]. Classical graph search algorithms are examples of common path planning algorithms [5], sampling-based algorithms [6], artificial potential field methods [7] and intelligent optimization algorithms [8]. Sampling-based path planning algorithms have drawn a lot of interest lately because of their exceptional performance in complicated situations and high-dimensional landscapes. Because of its ease of use and effectiveness, the rapid exploration random tree (RRT) algorithm has emerged as one of the traditional techniques in the field of path planning.

However, RRT algorithms still have deficiencies in terms of path quality, convergence speed, and adaptability to complex environments [9-12]. To overcome these limitations, researchers have proposed a variety of improvement methods. Gu Qian et al. proposed a combination of Gaussian probability bias strategy and swing search extension strategy, which can find the near-optimal path

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faster, but the application scenario of this algorithm is two-dimensional and is not applicable to multi-dimensional environments [13].

In addition, a large number of scholars have proposed other variant algorithms according to the RRT algorithm., Karaman et al proposed the RRT* algorithm, the algorithm increases the strategy of re-selecting the parent node and rewiring during the expansion process to gradually reduce the cost of generating paths, but relies on the uniform sampling of the configuration space, which leads to its slow convergence speed [14], JORDAN et al proposed the Bi-RRT* algorithm, which extends the growing tree from the initial position as well as the intended position respectively, and indicates that the two trees meet when the shortest distance between the two trees is less than a threshold, which reduces the influence of the target point position on the planning [15].

Aiming at the aforesaid issues, this study proposes a new improved RRT algorithm, which proposes a differential sampling mechanism for the issue of inadequate effectiveness of spatial sampling in 3D environment, and an adaptive step length method to dynamically modify the step length according to the environment's complexity in order to address the issue of the fixed-step environment's limited adaptability, and the progress of path planning. Aiming at the problem of poor path quality, a greedy pruning strategy and B-spline curve smoothing strategy are utilized to further improve the path quality by optimizing the selection and pruning process of the path nodes and adding path smoothing processing.

2. Robotic Arm Motion Model

The 6-DOF robotic arm used in this document was manufactured by Hefei Zhongke Shengu Company. The motion of the manipulator is based on positive and inverse kinematics. Positive kinematics involves providing six joint angles and solving for the end position of the manipulator. On the contrary, inverse kinematics involves solving for six joint angles. The DH method is utilized to create the linkage coordinate system of the 6-DOF manipulator arm as depicted in figure 1.

Table 1 displays the robotic arm's DH parameters, θ is the joint angle, d is the connecting link spacing, and a is the link length α is the initial angle of the link. The two linked links' transformation matrix can be obtained by two translations and rotations as shown in equation (1):

$$\begin{aligned} A_i &= \mathrm{Rot}(\theta_i)\mathrm{Trans}(d_i)\mathrm{Trans}(a_i)\mathrm{Rot}(\alpha_i) \\ &= \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned} \tag{1}$$

Where I=1, 2, ...,6, Rot () is the rotation matrix, Trans () is the translation matrix, is the cosine function and is the sine function.

DH parameter substitution in eq. (1), the transformation matrix between the two connecting links is available to us. DH parameter substitution in eq. (1), the transformation matrix between the two connecting links is available to us. Multiplying these six matrices, we can obtain the positive kinematic equations of the 6-DOF robotic arm as shown in eq. (2).

$$T = A_1 A_2 A_3 A_4 A_5 A = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

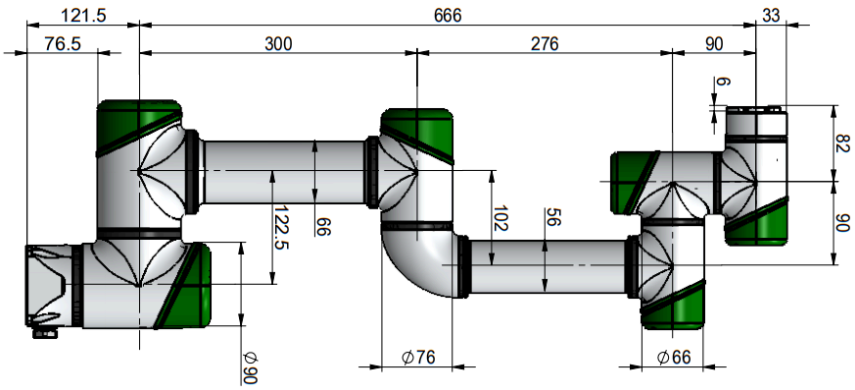


Figure 1. Parametric model of robotic arm.

Table 1. Robotic arm DH parameters.

link	initial angle α_{i-1} / (°)	joint angle θ_i / (°)	link length a_{i-1} / cm	connecting link spacing d_{i-1} / cm
1	0	θ_1	0	0
2	90	θ_2	0	122.5
3	0	θ_3	300	0
4	0	θ_4	275	102
5	90	θ_5	0	90
6	-90	θ_6	0	90

where p indicates the position of the robot arm's end and [n,o,a] indicates the end's attitude. robot manipulator.

3. Principle of Traditional RRT Algorithm

Random sampling is the foundation of the RRT algorithm, a path planning technique. The core of the algorithm lies in using the RRT algorithm to gradually build a search tree within the configuration space by random sampling, in order for it to progressively stretch from the beginning point to the destination. The RRT algorithm initially chooses a point in the configuration space at random to serve as a candidate random x_{rand} extension point for each iteration, and then searches for the closest existing node $x_{nearest}$ in the tree, and takes this node as the starting point, and expands it in the x_{rand} -direction in accordance with a predetermined fixed step size, thus generating a new node x_{new}. The RRT method looks for the nearest existing node $x_{nearest}$ in the tree before adding a new node. The RRT method adds nodes x_{new} to the search tree after determining whether the path between them conflicts with other obstacles; if not, it repeats the process. The RRT method progressively constructs a path in the configuration space that is free of collisions from beginning to end by repeatedly performing this operation.

4. RRT Algorithm Improvements

4.1. Differentiated Sampling Mechanism

Because sampling in space is random, some locations are sampled repeatedly, which reduces the random tree's efficiency in spatial sampling. This section proposes the differential sampling mechanism. The mechanism determines whether a randomly generated sample point is located in an explored region by calculating the distance from that sample point to every random tree node. The region is deemed fully explored if the distance is less than a specific threshold for differential sampling, in which case the sample point is discarded and resampled; on the other hand, if the

separation is greater than the threshold, the sample point is expanded into the random tree, preventing redundant node generation and duplicate sampling. The following equation (3) illustrates the differential node discrimination:

$$\begin{cases} \text{True}, & \min_{i=1,2,\ldots,n} \| x_{rand} - x_i \| > R_{\lim} \\ \text{False}, & \min_{i=1,2,\ldots,n} \| x_{rand} - x_i \| \le R_{\lim} \end{cases} \tag{3}$$

In equation (3) denotes i is the points in the generated tree, in figure 2, which also shows how the differentiated nodes developed. The random tree is deemed to have reached its current region when the distance between its nodes $x_2^{'}$ and point x_1 is less than the sampling threshold $R_{\lim}$; hence, $x_2^{'}$ is unnecessary to extend the random tree further. This circumstance is avoided by the differential sampling process, and the spread of random trees into unexplored areas of the site. The node formation procedure is likewise subject to this mechanism. The fact that no new nodes are created in the random tree's node domain through differential sampling—where the threshold $R_{\lim}$ value establishes the density of the random tree nodes—significantly enhances the quality of the random tree expansion.

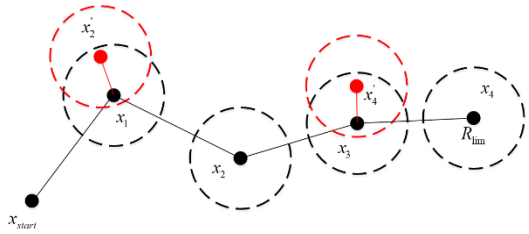


Figure 2. Differentiated node generation graph.

4.2. Adaptive Target Bias Strategy

The target bias sampling technique is suggested to improve the direction of sampling and accelerate the search speed because the RRT algorithm's sample unpredictability results in low search efficiency. Because the sampling process is explored by the RRT algorithm, when the density of obstacles is high, the free space in the map is less, the efficiency of random sampling is reduced, and the path planning is more difficult, at this time, increase the target bias probability that is to increase the threshold, which can guide the algorithm to the intended location to search faster, and reduce the ineffective sampling; when the density of obstacles is low, the efficiency of random sampling is elevated, and the path planning is relatively easy, at this time, it can be Appropriately reduce the target bias probability, lower the threshold, to maintain the algorithm's capacity to search and avoid premature convergence. Based on this characteristic, a heuristic target bias function is proposed to be determined according to the number of obstacles and the space occupation ratio as shown in equation (4) below.

$$x_{\text{rand}} = \begin{cases} x_{\text{random}}, & \text{if rand} > p \\ x_{\text{goal}}, & \text{otherwise} \end{cases} \tag{4}$$

$$p = \frac{1}{1+e^{-k(R_v)}} \tag{5}$$

$$R_v = \frac{V_{obs}}{V_{total}} \tag{6}$$

(4) where: x_{goal} indicates the intended location, x_{random} represents the random sampling point; p is the predetermined goal bias threshold; rand is the random number generation function; and denotes the final sampling point. k is the number of spatial obstacles, V_{obs} and V_{total} are the volume of obstacles and configuration space volume, respectively. R_v is the evaluation index of obstacle volume-space ratio, if the set bias threshold is smaller than the random value rand, Because the RRT algorithm's sample unpredictability leads to low search efficiency, to improve the direction of sampling and expedite the search, the target bias sampling approach is used.

4.3. Adaptive Step Length Mechanism

In a configuration space of the same size, the volume ratio of obstacles tends to determine the optimal sampling step size; the larger the volume occupied by obstacles, the smaller the sampling step size should be. To enhance the path quality, the sample step size should also be suitably decreased as the iteration goes on. Consequently, as seen in (7), we develop a variable sample step heuristic function by combining and the number of iterations. d_{step} is the size of the steps, S_{total} is the configuration space's boundary size. and i stand for the current number of iterations and $Iiter$ is the maximum number, respectively. $e^{\frac{-i}{Iiter}}$ is the nonlinear decay index factor, which is used to accelerate the step size decrease, and α is the maximum step size influence factor, which is used to restrict the step size.

$$d_{step} = (1/\alpha)e^{\frac{-i}{Iiter}}(S_{total} * (1 - R_v)) \tag{7}$$

4.4. Greedy Pruning Strategy and B-Spline Curve Optimization

Although the differential sampling mechanism improves the effectiveness of route planning in intricate settings, redundant nodes or paths still exist in random trees, and these redundant parts not only increase the computational burden, but also affect the smoothness and optimality of the paths. To further optimize the path quality and computational efficiency, this paper first introduces a greedy pruning strategy. This strategy aims to further simplify the random tree structure by removing redundant nodes, thus improving the optimality of paths.

Figure 3 illustrates the greedy pruning process. The resulting initial path is shown in figure 3(a), x_{start} is the index starting point, as shown in figure 3(b), collision occurs when the index target point is x_2, and the x_1 - x_2 line is divided into three equal parts to generate new nodes q_1 q_2, respectively, and the new node is connected to the node did not collide, connect x_{start} - q_1, while the node will be set as the starting point of the new index, as shown in figure 3(c), and continue to optimize to generate an optimized final path, as shown in figure 3(d).

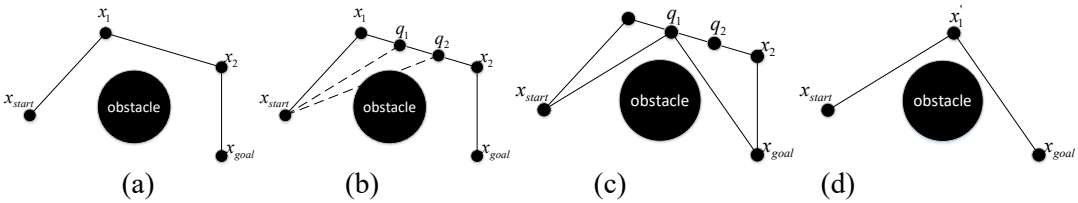


Figure 3. Schematic diagram of greedy pruning.

On this basis, all the superfluous points on the trail have been removed, but there are inflection points and it is not a smooth curve. The movement of the robotic arm in following the path will cause jitter, instability and other characteristics, so the path is smoothed using a B spline curve. The K times B spline curve equation is:

$$p(u) = \sum_{i=0}^{n} x_i N_{i,k}(u) \tag{8}$$

$$\begin{cases} N_{i,0}(u) = \begin{cases} 1, & u_i \le u \le u_{i+1} \\ 0, & else \end{cases} \\ N_{i,k}(u) = \dfrac{(u - u_i)}{(u_{i+k} - u_i)} N_{i-1,k}(u) + \dfrac{(u_{i+k+1} - u)}{(u_{i+k+1} - u_{i+1})} N_{i+1,k-1}(u) \\ \text{define}: \dfrac{0}{0} = 0 \end{cases} \tag{9}$$

The three times uniform spline curve is as follows, using the Cox-deBoor recursive formula defined in (9): where: K indicates the number of times B spline, $K+1$ indicates the curve order, i indicates the serial number of B spline, and x_i indicates the number of control points.

$$p_{0,3}(u) = x_0 \cdot N_{0,3}(u) + x_1 \cdot N_{1,3}(u) + x_2 \cdot N_{2,3}(u) + x_3 \cdot N_{3,3}(u) \quad (14) \quad u \in [0,1] \tag{10}$$

When the control points are given, the above equation can be used to find out the points of a three times B-style gentle curve. Figure 4 shows the example of the B-spline curve optimization.

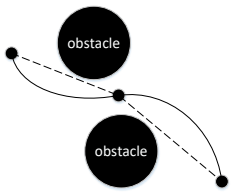


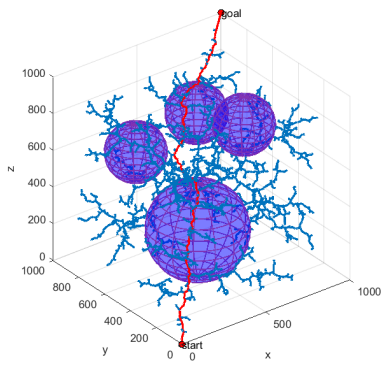
Figure 4. The example of the B-spline curve optimization.

5. Simulation Analysis and Experimental Verification

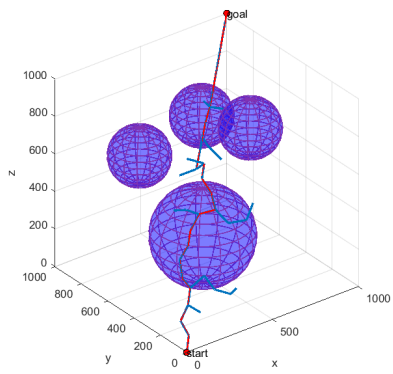
5.1. Validation of the Effectiveness of the Improvement Mechanism in a Three-dimensional Simulation Environment.

Several comparative tests are conducted in the MATLAB R2018a software to assess the effectiveness of the improvement technique in this work. The simulation experiments were conducted on an Intel(R)Core (TM) i5-9300 CPU running at 2.40GHz with 16GB of RAM. Different scenes in the 3D simulation environment are constructed, and taking into account the sampling algorithm's randomness, each group of experiments is executed for 50 times, figure 7 displays the simulation findings.

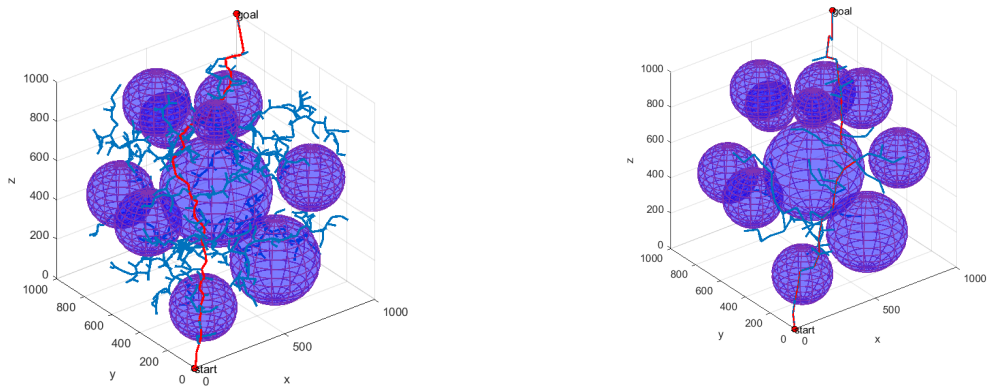
5.1.1. Validation of the Effectiveness of the Improvement Mechanism. Subsection combines the proposed improvement strategy with the RRT method, in order to confirm how well the combination of differential sampling, adaptive target bias and adaptive step size mechanism on the path search efficiency of the RRT algorithm, control experiments are set up in the simple obstacle environment and the complex obstacle environment, respectively, in order to validate the effectiveness. The simulation results are shown in figure 5.



(a)



(b)



(c) (d)

Figure 5. Plot of simulation results in different environments, map1 before improvements(a), map1 improved(b), map2 before improvements(c), and map2 improved (d).

Table 2. Improvement mechanism validation.

maps	algorithm	Path length	Planning time	Number of nodes
simple	RRT	2375	0.236	65
environment	updated algorithm	2015	0.042	35
complex	RRT	2392	0.553	60
environment	updated algorithm	2183	0.212	49

From figure 5(a) to figure 5(d) and table 2, it can be seen that in simple obstacle environments, the addition of differential sampling mechanism and adaptive step size mechanism produces fewer branches in the search tree, which can better pass through narrowly spaced as well as multi-obstacle regions, and reduces the path's length by 15.15%, reduces the planning time by 82.20%, and improves the effective node utilization rate by 46.15%; the above control experiments demonstrate that the enhanced approach can increase the algorithm's path-finding capabilities in surroundings with basic obstacles.. In the complex obstacle environment, compared with the original algorithm, there is a reduction of 8.86% in path length, a reduction of 61.66% in planning time, and an increase of 18.33% in effective node utilization; the aforementioned control tests demonstrate how the enhanced approach can improve the algorithm's path-finding capabilities in the intricate obstacle environment.

5.1.2. Validation of the Effectiveness of the Path Optimization Strategy. After generating the initial path, the improved method in this study improves the path. The path is then optimized following a greedy pruning strategy. A control experiment is set up in a straightforward obstacle setting to confirm the optimization's efficacy. Figure 6 displays the outcomes of the simulation.

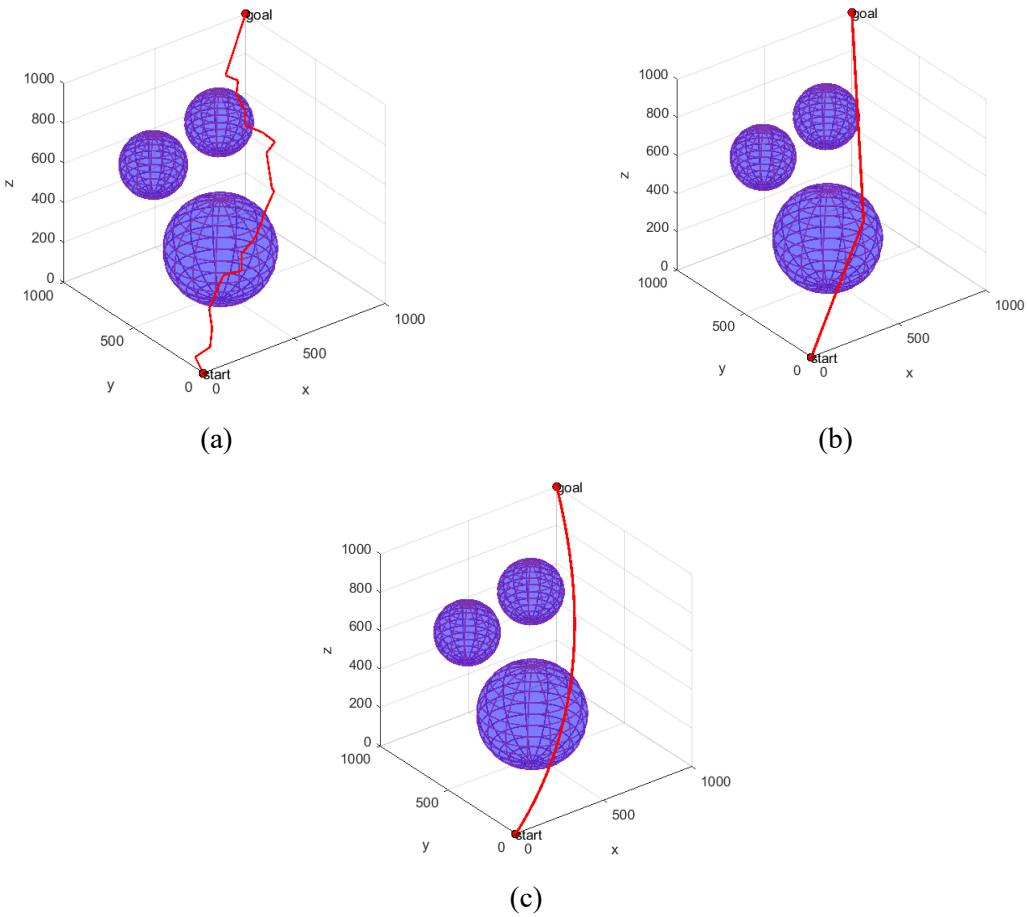


Figure 6. Path optimization effect diagram, before improvements(a), after pruning(b), and after curve smoothing(c).

As can be seen in figure 6, the greedy pruning technique is helpful in improving the quality of the path because it results in a much smaller number of nodes and a shorter average path length.5.1.3 Comparative analysis between this paper's algorithm and other algorithms

An improved analysis and comparison of the RRT method with the RRT* algorithm, BI-RRT* algorithm, and the original RRT methodology are presented in this subsection. Figure 7 shows the path planning graphs of different algorithms in a complex environment, and Table 3 compares the experimental data for search success rate, number of node points, path length, and search convergence time for each approach.

The enhanced RRT algorithm in this document is depicted in figure 7(d), which has a shorter search time, a shorter path length, and fewer nodes than the original RRT algorithm. The algorithm's planning time and path length are reduced by 57.89%. and 23.12%, respectively, and the RRT algorithm has a more complicated path and an average search success rate.In comparison to the RRT* method, which is seen in figure 7(b), the enhanced RRT cuts the algorithm's planning time and search path length by 20.28% and 20.01%, respectively. Figure 7(c) displays BI-RRT*. In comparison to the BI-RRT* algorithm, the enhanced RRT algorithm decreases the planning time and search path length by 15.47% and 13.70%, respectively. The four algorithms have relative success percentages of 92%, 94%, 93%, and 100%. It is evident from the comparison test results above that the enhanced RRT algorithm in this work may successfully raise the quality and efficiency of path planning.

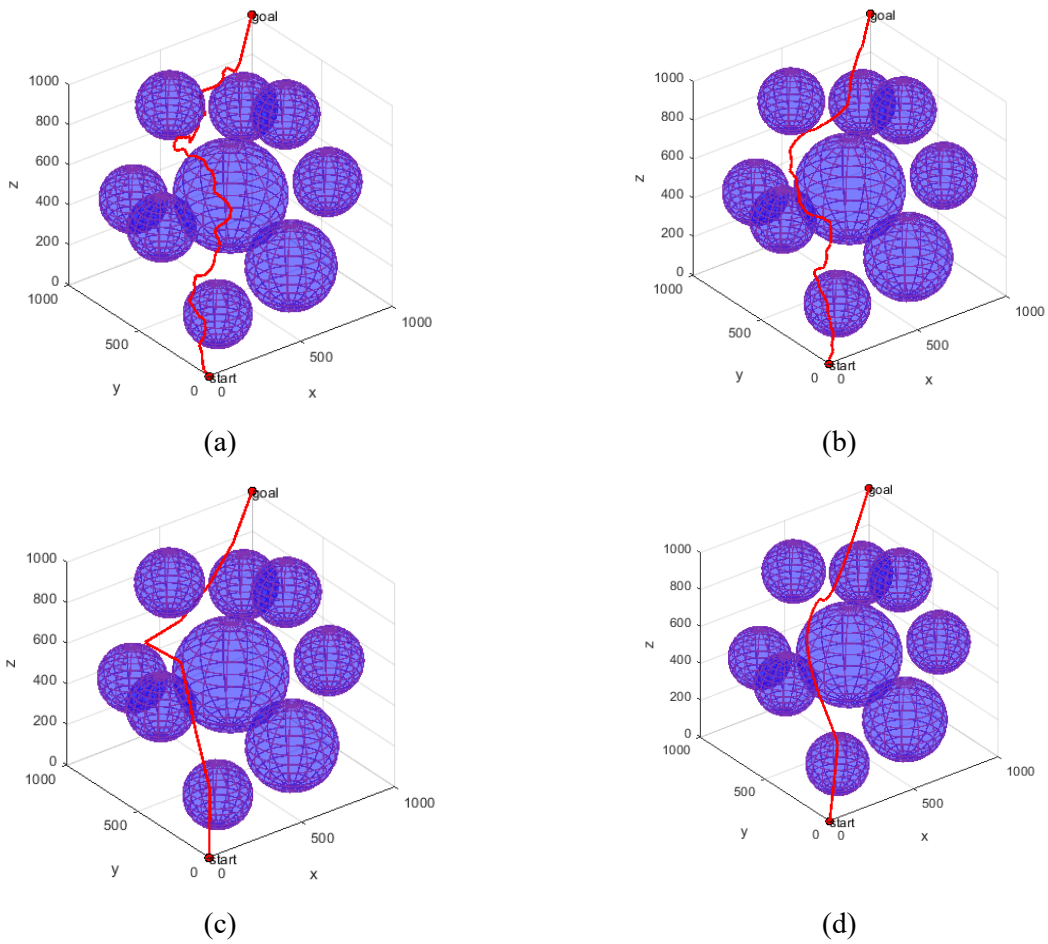


Figure 7. Simulation results of each algorithm, Original RRT (a), RRT* (b), BIRRT* (c), and the article's algorithm(d).

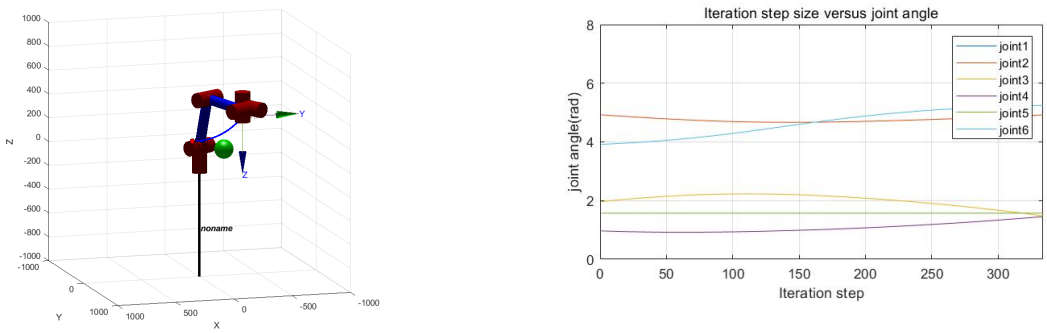
Table 3. Algorithm simulation average data.

algorithm	Path length	Planning time	Success Rate
RRT	2392	0.532	92
RRT*	2299	0.281	94
BIRRT*	2131	0.265	93
The algorithms in this paper	1839	0.224	100

5.2. Experimental Verification of the Robotic Arm

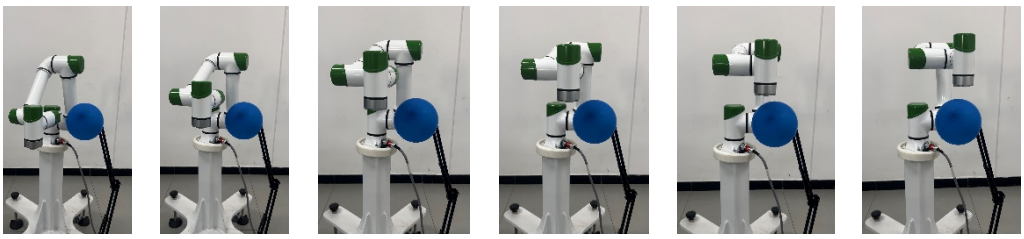
Figure 8 (a) depicts the robotic arm obstacle avoidance simulation experiment, where the robotic arm moves along a curve and successfully avoids the sphere obstacle. The first simulation using Matlab is conducted to ascertain the feasibility of the algorithm proposed in this study in practical applications. To confirm the stability of the robotic arm during movement, the various joints of the robotic arm during movement are analyzed. As illustrated in figure 8 (b), the changes in the robotic arm's angle during movement are analyzed to confirm the stability of the arm. The joints of the robotic arm are overly smooth during operation, and the angle changes of the joints are smooth curves. Second, to confirm the algorithm's viability, the experimental platform for the robotic arm is built, the Zhongke Shengu robotic arm is chosen, its URDF model is loaded, and the robotic arm executes the predetermined path points, and obstacles are positioned in accordance with the simulation

environment. In figure 9, the experimental diagram is shown. In accordance with the robotic arm's actual movement law, the simulation experiment demonstrates that the arm can successfully avoid obstacles when in use.



(a) (b)

Figure 8. Matlab simulation diagram, robotic arm obstacle avoidance simulation (a), and iteration step size versus joint angles (b).



(a) (b) (c) (d) (e) (f)

Figure 9. Physical experiment diagram.

6. Conclusion

In order to calculate the obstacle avoidance path for robotic arms, an enhanced RRT algorithm is suggested. On the basis of robotic arm kinematics, the efficiency of collision detection is improved by introducing an envelope to simplify the robotic arm model; on the basis of the RRT algorithm, firstly, a differential sample technique is suggested to enhance the sampling efficiency; secondly, an adaptive target bias and adaptive step length mechanism on the basis of spatial occupancy of the obstacles and the progress of the path planning is proposed to improve the adaptability of the target bias and the step length to the environment; and finally, a greedy pruning strategy and a B-spline curve smoothing strategy are used to optimize the initial path pruning and path smoothing to improve the path quality. greedy pruning strategy and B-spline curve smoothing strategy to optimize the initial path pruning and path smoothing to enhance the quality of the path. The findings indicate that the improved algorithm can successfully raise the associated performance. Applying the improved algorithm to the physical platform further demonstrates the algorithm's viability and efficacy. Future research will concentrate on improving of the obstacle model description and further improve the algorithm to cope with more complex obstacle avoidance environments.

Acknowledgments

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